USN | | | | | | | | | | | [22BSMA21]

SECOND SEMESTER B.Tech EXAMINATION, JULY - 2023
TRANSFORMS AND PARTIAL DIFFERENTIAL EQUATIONS

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**05X02=10**
1.	**a.**	Evaluate $\int_0^1 \int_1^1 xy\, dydx$	L-2	CO1	**02**
	b.	Find the Fourier co-efficient a_0 for the function $f(x) = 1 - x^2$ in $(-1,1)$.	L-2	CO2	**02**
	c.	Write the suitable formula to find the Fourier sine transform of $f(t) = \begin{cases} 1, 0 \le t < a \\ 0, t \ge a \end{cases}$ by considering the given limits.	L-1	CO3	**02**
	d.	Find the Z-transform of $(\cos\theta + i\sin\theta)^n$.	L-2	CO4	**02**
	e.	Solve $\frac{\partial^2 u}{\partial x^2} = x + y$ by direct integration.	L-3	CO5	**02**
		PART - B			
2.	**a.**	Write Matlab code to find the double integral of any function.	L-3	CO1	**05**
		OR			
	b.	Write Matlab code to find the inverse Fourier transform of any function.	L-3	CO3	**05**
3.	**a.**	Evaluate $\int_0^1 \int_x^{\sqrt{x}} (x^2 + y^2) dydx$.	L-3	CO1	**05**
		OR			
	b.	Evaluate $\iint_R ydxdy$ where $R$is the region bounded by the first quadrant of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.	L-3	CO1	**05**
4.	**a.**	Obtain the Fourier cosine series expansion of $f(x) = \frac{1}{2}(\pi - x)$ in $(0, \pi)$.	L-3	CO2	**05**
		OR			
	b.	Expand $f(x) = x(\pi - x)$ as half range sine series over the interval $(0, \pi)$.	L-3	CO2	**05**
5.	**a.**	Find the Fourier transform of the function $f(t) = e^{-a\|t\|}$ where $a > 0$.	L-3	CO3	**05**
		OR			
	b.	Find the Fourier sine transform of $f(t) = \frac{e^{-at}}{t}, a > 0$.	L-3	CO3	**05**

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

6.	**a.**	Find the z-transform of $2n + \sin\left(\frac{n\pi}{4}\right) + 1$.	L-3	CO4	**05**
		OR			
	b.	Find the inverse z-transform of $\frac{z}{(z-1)(z-2)}$.	L-3	CO4	**05**
7.	**a.**	Solve $\frac{\partial^2 z}{\partial x \partial y} = \sin x \sin y$ given that $\frac{\partial z}{\partial y} = -2\sin y$ when $x = 0$ and $z = 0$ when y is an odd multiple of $\frac{\pi}{2}$ by direct integration method.	L-3	CO5	**05**
		OR			
	b.	Solve $(y - z)p + (z - x)q = (x - y)$.	L-3	CO5	**05**
		PART - C			
8.	**a.**	Write Matlab code to find the Fourier series expansion of any function.	L-3	CO2	**10**
		OR			
	b.	Find the z-transform of the function $\left(\frac{1}{n+1}\right)$. Also write the Matlab code for the same.	L-3	CO4	**10**
9.	**a.**	Evaluate $\int_0^3 \int_1^{\sqrt{4-y}} (x + y)dxdy$ by changing the order of integration.	L-3	CO1	**10**
		OR			
	b.	Evaluate $\int_0^a \int_y^a \frac{x^2}{\sqrt{x^2+y^2}} dxdy$ by changing to polar coordinates.	L-3	CO1	**10**
10.	**a.**	Obtain the Fourier series of y up to the second harmonics for the following values.	L-3	CO2	**10**

x^0	45	90	135	180	225	270	315	360
y	4	3.8	2.4	2	-1.5	0	2.8	3.4

		OR			
	b.	Find the Fourier series expansion of $f(x) = \begin{cases} -k, -\pi < x < 0 \\ k, 0 < x < \pi \end{cases}$ where k is constant and hence deduce that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots$	L-3	CO2	**10**
11.	**a.**	Find the Fourier transform of $f(x) = \begin{cases} 1 - x^2, \lvert x\rvert < 1 \\ 0, \lvert x\rvert \geq 1 \end{cases}$ and hence evaluate $\int_0^\infty \frac{x\cos x - \sin x}{x^3} dx$.	L-3	CO3	**10**
		OR			
	b.	Find the cosine transform of $f(t) = e^{-at}, a > 0$ and hence evaluate $\int_0^\infty \frac{\cos kx}{x^2+a^2} dx$	L-3	CO3	**10**

12.	**a.**	Obtain the Z-transform of $\cos n\theta$ and $\sin n\theta$. Hence deduce z-transform of the following $i)\ k^n \cos n\theta,\ \ ii)\ k^n \sin n\theta,\ \ iii)\ e^{-an} \cos n\theta,\ iv)\ e^{-an} \sin n\theta.$	L-3	CO4	**10**
		OR			
	b.	Solve the difference equation $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$ with $y_0 = y_1 = 0$ using Z-transform.	L-3	CO4	**10**
13.	**a.**	Find the solution of one-dimensional heat equation by the method of separation of variables.	L-3	CO6	**10**
		OR			
	b.	Solve $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} = 0$, by the method of separation of variables.	L-3	CO6	**10**